



From Docking to Dynamics

Preparing Protein–Ligand Systems with CHARMM-GUI for GROMACS Simulations

Presented by:
Donia Gamal Youssef Soliman

Assistant lecturer of Biophysics
Medical Biotechnology program
Badr University in Cairo

Workflow Overview

1. Add Chain ID (alter sele,chain='B')
2. Check Bonds
3. Add Hydrogens (h_add)
4. Save system as .pdb
5. Generate ligand parameters via CGenFF
6. Adjust PDB — Chain ID & resn using lig.cgenff.mol2
7. Save in original atom order
8. Split .str → topology.rtf + parameter.prm
9. Rename ligand in topology
10. Launch CHARMM-GUI
11. ⚠ Carefully inspect and define any disulfide bonds in your protein

